Chapter 2: Neuron & Glia.

Nerve Cell
- neuron 神经元 85 billion
  - processing of information ("nerve impulse" 电脉冲)
    - sensation
    - intergration
    - movement
- glia → glue 胶质细胞 85 billion
  - insulation 绝缘
  - nourishment
  - support

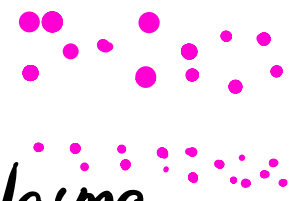
BBB Blood-Brain-Barrier

Cellular neuroscience

- microscope
- Brain tissue is soft. "harden"
  - 4% 多聚甲醛 福尔马林 formaldehyde
  - rapid protein denature → 切片机 rapid fixation.
- Slice into pieces. microtome
  - thickness "50 ~ 100um"
- Colorize / Pigmentation

"the gain in brain is mainly the stain"

Franz Nissl → Nissl Stain.



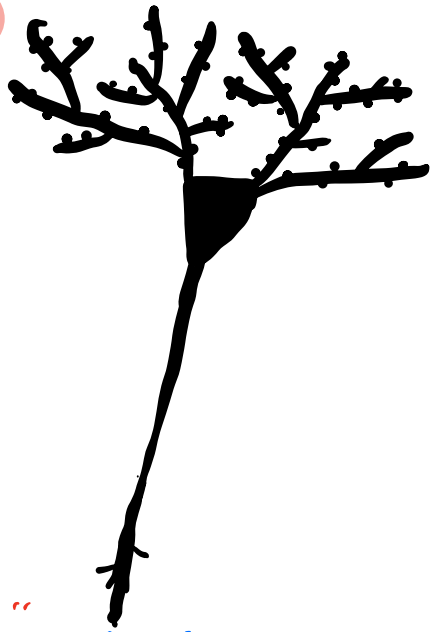
- label of cell soma [nucleus] + protoplasma.
- recognize neuron. no label on glia.
- cytoarchitecture → arrange of nerve cell. lamination

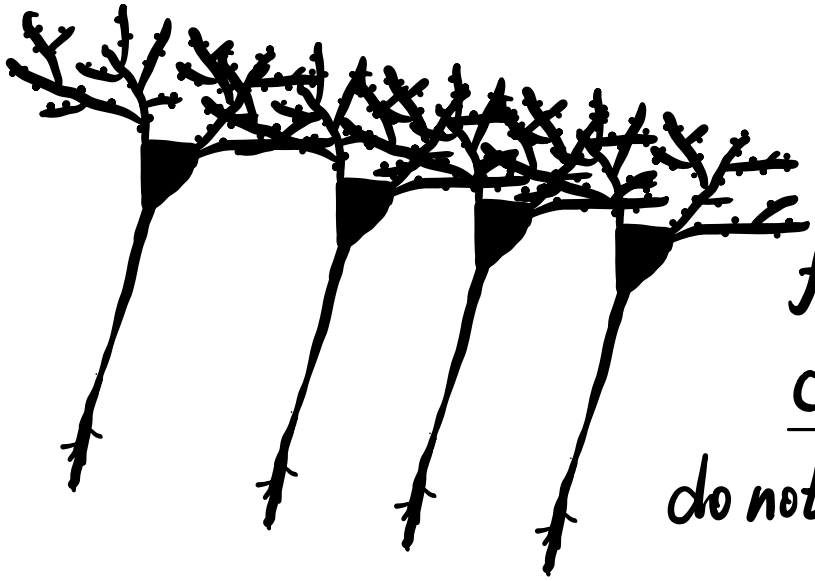
FSY

Camillo Golgi

1873. Golgi Stain

- dark precipitation, silver chromate
- only ~5% ~~neuron~~ nerve cell been labeled





"falsified" 网状结构

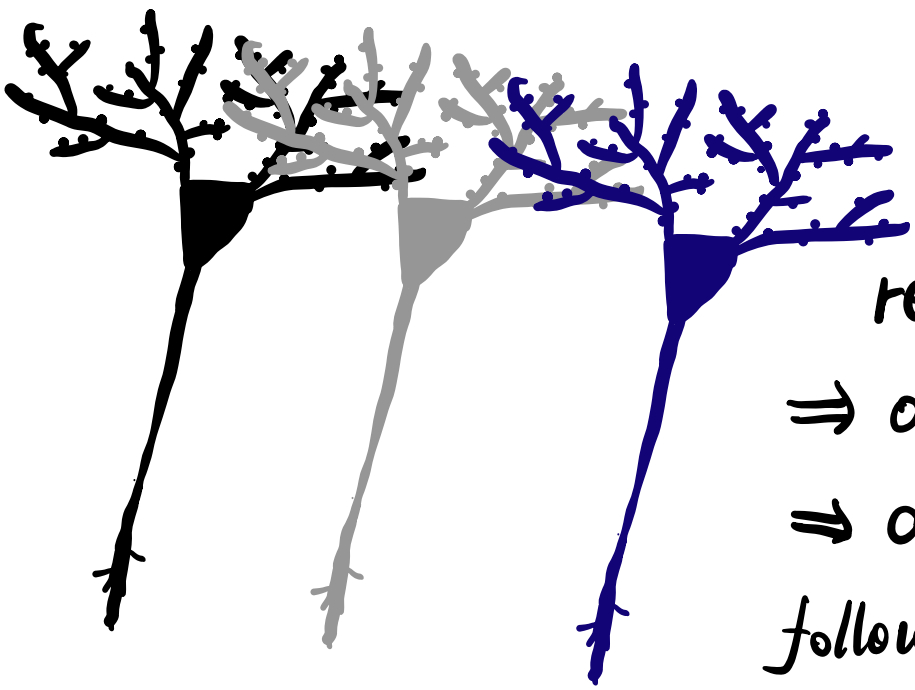
forming reticulum structure

capillary, artery, vein

do not follow cell theory.

cell is NOT smallest unit of life

Cajal. 1888. modified Golgi Stain Method.



reticulum structure

⇒ overlapped neuron

⇒ discreted neuron } neural doctrine

follow cell theory.

1906 Nobel Prize



▶ FIGURE 2.7
Neurites in contact, not continuity. These neurites were reconstructed from a series of images made using an electron microscope. The axon (colored yellow) is in contact with a dendrite (colored blue). (Source: Courtesy of Dr. Sebastian Seung, Princeton University, and Kris Krug, Pop Tech.)

Cell Structure

nucleus: DNA, control all cellular activity.

rER: synthesis protein for secretion.

membrane protein. e.g. ion channel, receptor of NT

neurotransmitter (NT)

mitochondria: aerobic respiration, ATP.

Na^+ - K^+ - Pump. vesicle transport

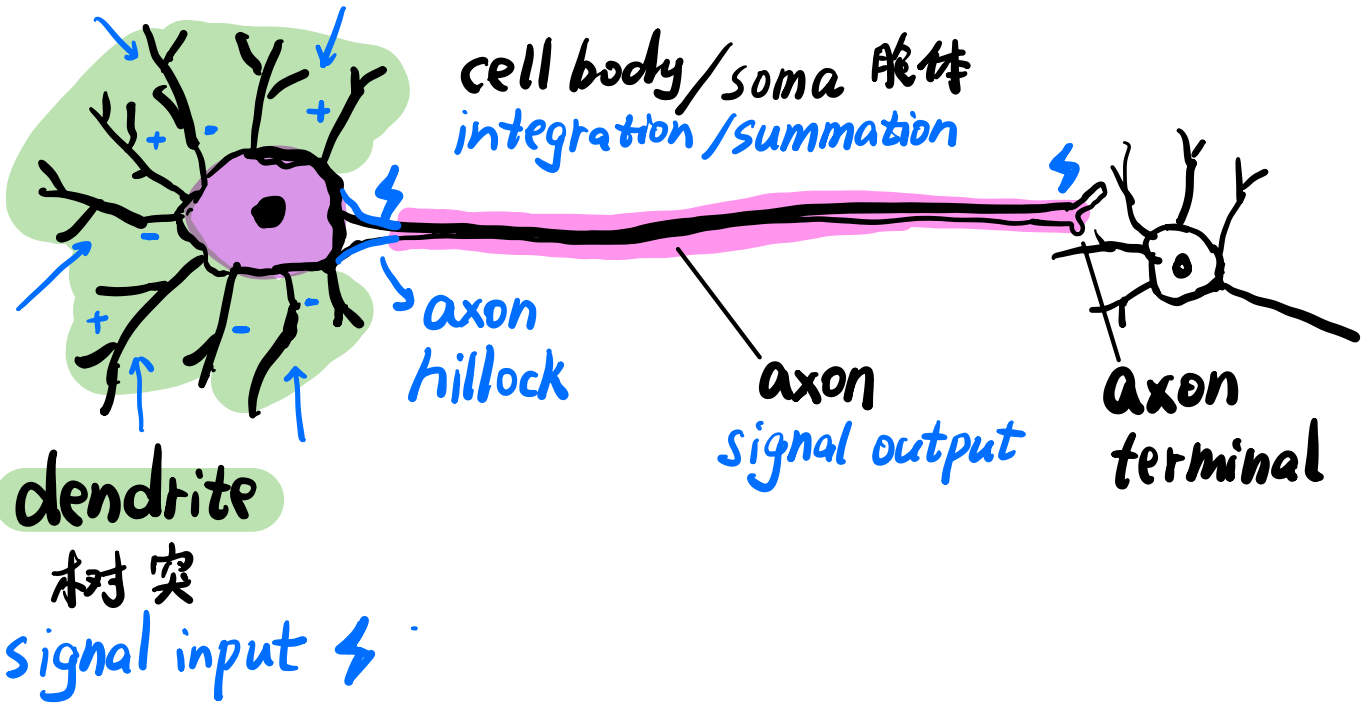
cytoskeleton: support cell shape.

MT. microtubule — vesicle transport

NF, neurofilament — **"tau" protein** → NFT → "Alzheimer's Disease"

MF, microfilament

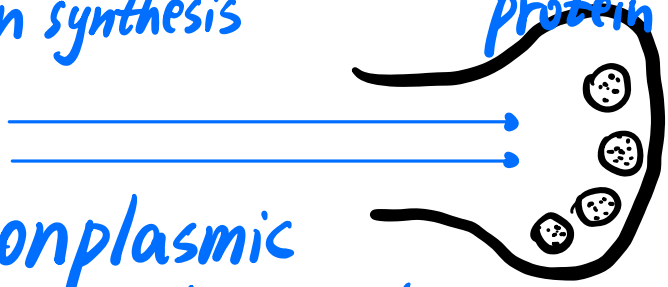
structure of a classical neuron.



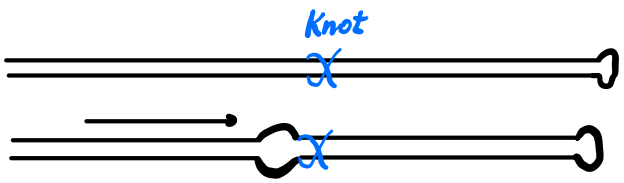
	Soma	Axon terminal
rER	✓	✗
free ribosome	✓	✗
mitochondria	✓	✓
vesicle	✗	✓

protein synthesis

protein usage.



axonplasmic transport



blocking material transfer.

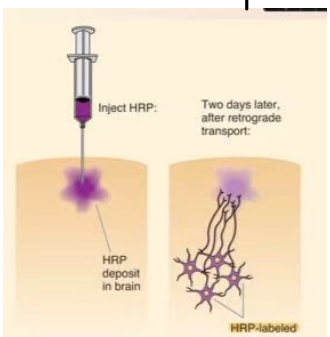
Radioactive labelled amino acid

anterograde	soma → axon terminal	Kinesin (e.g. vesicle transfer) Kick
retrograde	axon terminal → soma	Dynein Drag (e.g. Rabies Virus 狂犬病病毒)

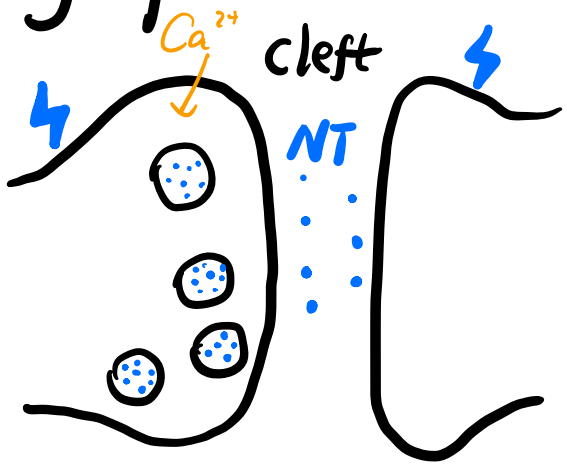
AKRD

"HRP label"

"terminal metabolite"



Synapse ≈ Chemical Synapse.



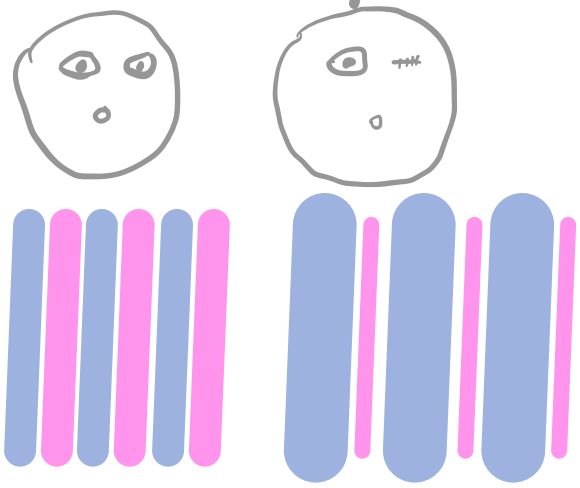
electrical signal
↓
chemical signal (NT)
↓
electrical signal

pre-synapse
axon terminal

post-synapse
dendrite / soma / axon

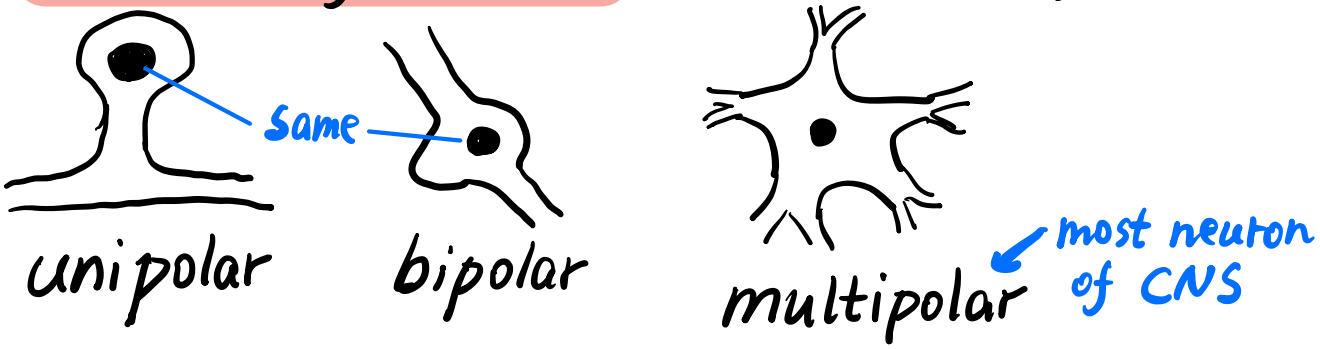
- memory
- learning / plasticity 可塑性.

monocular deprivation

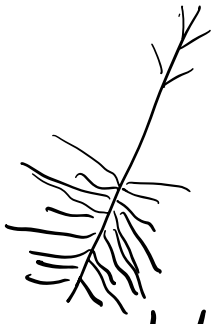


Classify of neurons.

- number of neurite [dendrite & axon]



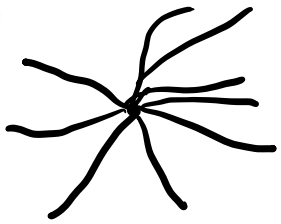
· shape of dendrites





pyramidal

most are spiny



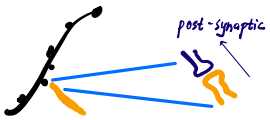
stellate

"Stella Lou"



"spiny & aspinous"

· dendritic spine ?



Spiny



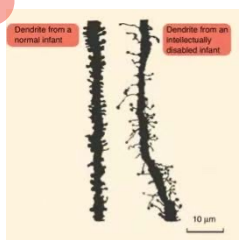
Aspinous



number of dendritic spine

→ intellectual level.

e.g. PKU, 21st trisomy



Length of axon

Golgi Type I	longer axon	projection
Golgi Type II	shorter axon	local

Connection

- Sensory neuron e.g. retina, olfactory epithelium,
- Interneuron
- motor neuron e.g. muscle & gland

Specific Genes.

GFP [Green Fluorescent Protein]

2008 Nobel Prize

GFP + specific promotor / control element.

- structural imaging (no signal / change of signal)

GCamP reveal Ca^{2+} signal → activity.

- functional imaging.

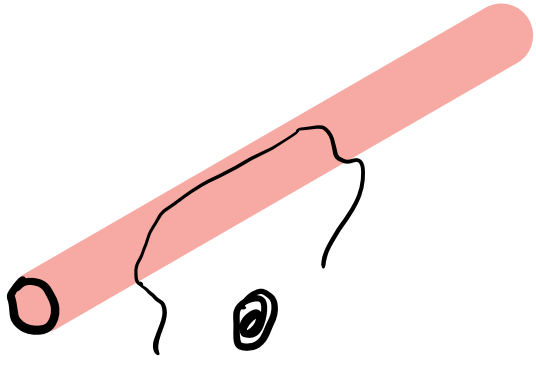
electrical signal

+ excitatory

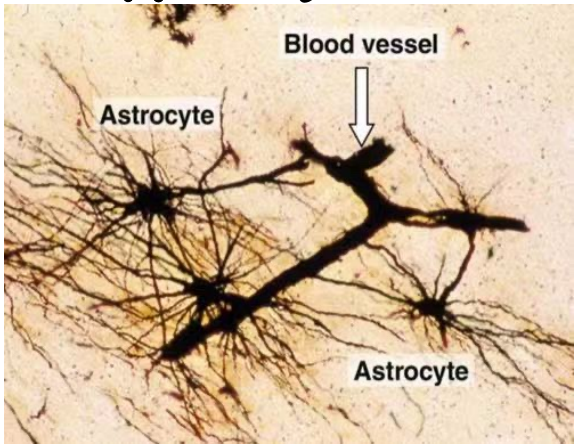
- inhibitory

Glia Cell Insulation & Nourishment

astrocyte 星形胶质细胞 *aster



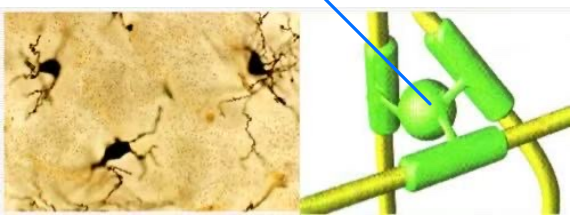
- Blood Brain Barrier, BBB
- fill in most area of brain
  affect neurite formation
- regulate extracellular "NT & K^+ ions"
- trigger blood flow. "fMRI"
- largest portion of glia cell.



Myelination Glia. Insulation → speedup nerve impulse

- Oligodendroglia cell 少突胶质细胞

CNS





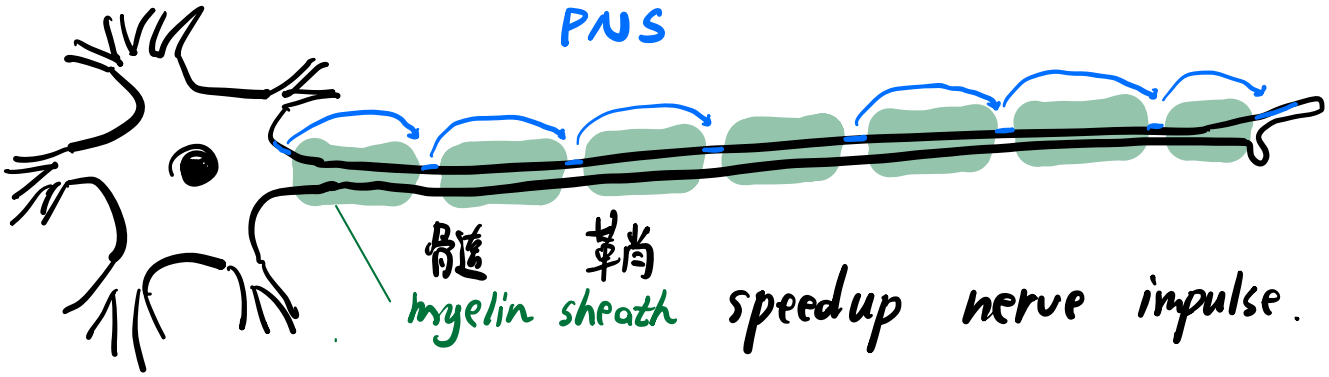
CNS PNS

COPS

Oligo~ Schwann

- Schwann Cell 施旺细胞.

PNS



髓鞘 myelin sheath speedup nerve impulse.

- Microglia 小胶质细胞

immune response → phagocyte similar to

phototaxi 向光性

chemotaxi 趋化性

taxi → movement

- Ependymal 室管膜 cell: lining tissue of ventricle
  - brain development
  - secretion of CSF